Data Science Professional Course Outline

Course Overview

A comprehensive program covering programming, data management, analysis, visualization, and machine learning with real-world applications.

Module 1: Introduction to Data Science

1.1 What is Data Science?

- The data science lifecycle
- Roles and responsibilities of data scientists
- Industry applications and career paths
- Tools and technologies overview

1.2 Setting Up Your Environment

- Installing Python (Anaconda distribution)
- Installing R and RStudio
- Jupyter Notebooks and IDE setup
- Version control with Git basics

Module 2: Python Programming Fundamentals

2.1 Python Basics

- Variables, data types, and operators
- Control structures (if/else, loops)
- Functions and modules
- Error handling and debugging

2.2 Data Structures

- Lists, tuples, sets, and dictionaries
- List comprehensions
- Working with strings
- File I/O operations

2.3 Essential Libraries

- NumPy for numerical computing
- Pandas for data manipulation
- Introduction to scikit-learn
- Working with dates and times

Project: Build a data processing pipeline for CSV files

Module 3: R Programming Fundamentals

3.1 R Basics

- R syntax and data types
- Vectors, matrices, and data frames
- Control structures and functions
- Packages and libraries

3.2 Data Manipulation in R

- dplyr for data transformation
- tidyr for data tidying
- Data import/export
- Working with factors and dates

3.3 R vs Python

- When to use each language
- Comparing syntax and capabilities
- Integrating R and Python workflows

Project: Statistical analysis report using R

Module 4: SQL and Database Management

4.1 Database Fundamentals

- Relational database concepts
- Database design and normalization
- Entity-relationship diagrams
- Installing and setting up MySQL

4.2 SQL Basics

- SELECT, FROM, WHERE clauses
- Filtering and sorting data
- Aggregate functions (COUNT, SUM, AVG, etc.)
- GROUP BY and HAVING

4.3 Advanced SQL

- JOIN operations (INNER, LEFT, RIGHT, FULL)
- Subqueries and nested queries
- UNION and set operations
- Window functions

4.4 Database Management

- Creating and modifying tables
- INSERT, UPDATE, DELETE operations
- Indexes and query optimization
- Connecting Python/R to MySQL databases

Project: Design and query a multi-table database system

Module 5: Data Cleaning and Preparation

5.1 Understanding Data Quality

- Types of data issues
- Missing data patterns
- Outliers and anomalies
- Data validation techniques

5.2 Data Cleaning Techniques

- Handling missing values (imputation strategies)
- Removing duplicates
- Data type conversions
- String cleaning and standardization

5.3 Data Transformation

- Feature scaling and normalization
- Encoding categorical variables

- Binning and discretization
- Creating derived features

5.4 Data Integration

- Merging and joining datasets
- Concatenating data
- Reshaping data (pivot, melt)
- Handling time series data

Project: Clean and prepare a messy real-world dataset

Module 6: Exploratory Data Analysis

6.1 Statistical Foundations

- Descriptive statistics
- Probability distributions
- Hypothesis testing basics
- Correlation and covariance

6.2 Data Exploration Techniques

- Univariate analysis
- Bivariate analysis
- Multivariate analysis
- Identifying patterns and relationships

6.3 Data Profiling

- Data summarization
- Distribution analysis
- Detecting outliers
- Feature importance assessment

Project: Complete EDA report with insights and recommendations

---

Module 7: Data Visualization

7.1 Visualization Principles

- Design principles and best practices
- Choosing the right chart type
- Color theory and accessibility
- Storytelling with data

7.2 Python Visualization

- Matplotlib fundamentals
- Seaborn for statistical plots
- Plotly for interactive visualizations
- Pandas plotting capabilities

7.3 R Visualization

- Base R plotting
- ggplot2 grammar of graphics
- Interactive plots with plotly
- Creating dashboards with Shiny

7.4 Advanced Visualizations

- Heatmaps and correlation matrices
- Geographic visualizations
- Network graphs
- Time series plots

Project: Create an interactive dashboard for business metrics

---

Module 8: Machine Learning Fundamentals

8.1 Introduction to Machine Learning

- Supervised vs unsupervised learning
- Training, validation, and test sets
- Overfitting and underfitting
- Bias-variance tradeoff

8.2 Model Evaluation

- Cross-validation techniques
- Evaluation metrics (accuracy, precision, recall, F1)
- Confusion matrices
- ROC curves and AUC

8.3 Feature Engineering

- Feature selection methods
- Dimensionality reduction (PCA)
- Feature importance
- Handling imbalanced data

Project: Build evaluation framework for ML models

Module 9: Regression Algorithms

9.1 Linear Regression

- Simple and multiple linear regression
- Assumptions and diagnostics
- Regularization (Ridge, Lasso, Elastic Net)
- Polynomial regression

9.2 Advanced Regression

- Decision tree regression
- Random forest regression
- Gradient boosting (XGBoost, LightGBM)
- Support vector regression

9.3 Time Series Forecasting

- ARIMA models
- Exponential smoothing
- Prophet for forecasting
- Evaluating forecast accuracy

Project: Predict housing prices or sales forecasting

Module 10: Classification Algorithms

10.1 Basic Classification

- Logistic regression
- Decision trees for classification
- K-nearest neighbors (KNN)

- Naive Bayes

10.2 Advanced Classification

- Random forests
- Gradient boosting classifiers
- Support vector machines (SVM)
- Neural networks introduction

10.3 Multi-class and Multi-label

- One-vs-rest and one-vs-one strategies
- Multi-output classification
- Ensemble methods
- Calibration of probabilities

Project: Customer churn prediction model

Module 11: Unsupervised Learning

11.1 Clustering

- K-means clustering
- Hierarchical clustering
- DBSCAN
- Evaluating clustering quality

11.2 Dimensionality Reduction

- Principal Component Analysis (PCA)
- t-SNE visualization
- UMAP
- Autoencoders introduction

11.3 Association Rules

- Market basket analysis
- Apriori algorithm
- FP-growth algorithm
- Rule evaluation metrics

Project: Customer segmentation analysis

Module 12: Real-World Application - Fraud Detection

12.1 Understanding Fraud Detection

- Types of fraud
- Challenges in fraud detection
- Class imbalance problems
- Feature engineering for fraud

12.2 Building Fraud Detection Models

- Anomaly detection techniques
- Supervised classification approach
- Real-time scoring systems
- Model interpretability

12.3 Case Study

- Financial transaction fraud
- Credit card fraud detection
- Insurance claim fraud
- Model deployment considerations

Project: End-to-end fraud detection system

Module 13: Real-World Application - Recommendation Systems

13.1 Recommendation Fundamentals

- Types of recommendation systems
- Collaborative filtering
- Content-based filtering
- Hybrid approaches

13.2 Building Recommenders

- User-based collaborative filtering
- Item-based collaborative filtering
- Matrix factorization

- Deep learning for recommendations

13.3 Case Study

- Movie/product recommendations
- Personalization strategies
- Cold start problem
- Evaluation metrics (RMSE, MAE, precision@k)

Project: Build a movie or product recommendation engine

Module 14: Model Deployment and Production

14.1 Model Serialization

- Saving and loading models
- Pickle and joblib
- Model versioning
- Reproducibility best practices

14.2 API Development

- Flask/FastAPI basics
- Creating REST APIs
- Input validation
- Error handling

14.3 Deployment Strategies

- Cloud platforms overview (AWS, Azure, GCP)
- Containerization with Docker
- Model monitoring
- A/B testing

Project: Deploy ML model as a web service

Module 15: Capstone Project

15.1 Project Planning

- Problem definition and scope

- Data collection strategy
- Success metrics definition
- Project timeline

15.2 Execution

- Data acquisition and cleaning
- Exploratory analysis
- Model development and tuning
- Results documentation

15.3 Presentation

- Technical report writing
- Creating presentations
- Communicating results to stakeholders
- Building a portfolio piece

Final Deliverable: Complete data science project with documentation and presentation

---

Course Resources

Required Tools

- Python 3.8+, R 4.0+
- MySQL 8.0+
- Jupyter Notebook
- Git and GitHub

Key Libraries

- **Python:** pandas, numpy, scikit-learn, matplotlib, seaborn, plotly
- **R:** dplyr, ggplot2, caret, tidyverse

Supplementary Materials

- Dataset repositories (Kaggle, UCI ML Repository)
- Research papers and case studies
- Industry best practices documentation
- Community forums and support groups

Assessment Structure

- **Weekly Assignments:** 30%
- **Module Projects:** 30%
- **Mid-term Exam:** 15%
- **Capstone Project:** 25%

Prerequisites

- Basic mathematics (algebra, statistics)
- Computer literacy